

Smart Route Optimization System (Readme)

Overview

This project finds the shortest path between a source and destination using **Dijkstra's Algorithm (Greedy approach)**.

Algorithm Used

- Dijkstra's Algorithm
- Priority Queue (Min Heap)

Features

- Finds shortest path between two cities
- Displays distance and route
- Handles no-path cases
- Efficient implementation

How to Run

```
g++ src/main.cpp
```

```
./a.out
```

Sample Input

```
4
```

```
4
```

```
0 1 4
```

```
0 2 2
```

```
1 3 5
```

```
2 3 1
```

```
0
```

```
3
```

Output

Shortest path from 0 to 3:

Distance: 3

Path: 0 2 3

Complexity

- Time: $O(E \log V)$
- Space: $O(V)$

Limitations

- Does not work with negative weights